

Chapter 7

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Q: What are the edge labels in the suffix tree for "panamabananas\$"?

A: The suffix tree for "panamabananas\$" shown in the main text contains 17 edges with the following labels (note that different edges may have the same labels):

```
$
a
bananas$
mabananas$
na
mabananas$
nanas$
s$
s$
bananas$
mabananas$
na
mabananas$
nas$
s$
panamabananas$
s$
```

Q: How can I construct a suffix tree in linear time?

A: Suffix trees were introduced by Peter Weiner in 1973. However, the original linear-time algorithm for building the suffix tree was extremely complex. Although the Weiner algorithm was greatly simplified by Esko Ukkonen in 1995, it is still non-trivial. Check out the following excellent StackOverflow post by [Johannes Goller](#) if you are interested in seeing a full explanation:

<http://stackoverflow.com/a/9513423>

Q: Is it possible to construct the Burrows-Wheeler Transform in linear time?

A: Our naive approach to constructing $BWT(Text)$ requires construction the matrix $M(Text)$ of all cyclic rotations, which requires $O(|Text|^2)$ time and space. However, there exist algorithms constructing $BWT(Text)$ in linear time. One such algorithm first constructs the suffix array of $Text$ in linear time and then uses this suffix array to construct $BWT(Text)$.

Q: Can reads that "fall off the edges of the text" form approximate matches?

A: For example, does $Pattern = "T T A C T G"$ match $Text = "A C T G C T G C T G"$ with $d = 2$ mismatches?

Not according to the statement of the Multiple Approximate Pattern Matching Problem, since there is no *starting position* in *Text* where *Pattern* appears as a substring with at most d mismatches.

Q: How does the repeated triplet "CAG" affect the severity of Huntington's disease?

A: Huntington's disease is a rare genetic disease in that it is attributable to a single gene, called *Huntingtin*. This gene includes a trinucleotide repeat "...CAGCAGCAG..." that varies in length. Individuals with fewer than 26 copies of "CAG" in their *Huntingtin* gene are classified as unaffected by Huntington's disease, whereas individuals with more than 35 copies carry a large risk of the disease, and individuals with more than 40 copies will be afflicted. Moreover, an unaffected person can pass the disease to a child if the normal gene mutates and increases the repeat length. The reason why many repeated copies of "CAG" in *Huntingtin* leads to disease is that this gene produces a protein with many copies of glutamine ("CAG" codes for glutamine), which increases the decay rate of neurons.

Q: How can I modify the approximate pattern matching with the Burrows-Wheeler transform to account for patterns whose last symbols do not appear in text?

A: The algorithm illustrated in the epilogue would fail to find an approximate match of "nad" because the final symbol of "nad" does not appear in "panamabananas\$". To address this complication, we can modify the algorithm for finding a pattern of length m with up to k mismatches as follows.

We first run the algorithm described in the main text to find all approximate instances of a *Pattern* of length k against *Text*. However, this algorithm does not actually find all approximate matches of *Pattern* – since we do not allow mismatched strings in the early stages of **BetterBWMatching**, we miss those matches where the last letter of *Pattern* does not match *Text*. To fix this shortcoming, we can simply find all locations in *Text* where the prefix of *Pattern* of length $k - 1$ has $d - 1$ mismatches.. Yet this algorithm fails to find matches where the last two letters of *Pattern* do not match *Text*. Thus, we need to run the algorithm again, finding all locations in *Text* where the prefix of *Pattern* of length $k - 2$ has $d - 2$ mismatches. We then find all locations in *Text* where the prefix of *Pattern* of length $k - 3$ occurs with $d - 3$ mismatches, and so on, finally finding all locations in *Text* where the prefix of *Pattern* of length $k - d$ occurs exactly.

Created Mon 2 Feb 2015 9:43 PM PST

Last Modified Mon 16 Feb 2015 12:55 AM PST